4	**(a)**	work done moving unit positive charge		M1	
		from infinity (to the point)		A1	[2]
	(b)	(gain in) kinetic energy = change in potential energy		B1	
		$\frac{1}{2}mv^2 = qV$ leading to $v = (2Vq/m)^{\frac{1}{2}}$		B1	[2]
	(c)	*either*	$(2.5 \times 10^5)^2 = 2 \times V \times 9.58 \times 10^7$	C1	
			$V = 330\,\text{V}$	M1	
			this is less than 470 V and so 'no'	A1	[3]
		or	$v = (2 \times 470 \times 9.58 \times 10^7)$	(C1)	
			$v = 3.0 \times 10^5\,\text{m}\,\text{s}^{-1}$	(M1)	
			this is greater than $2.5 \times 10^5\,\text{m}\,\text{s}^{-1}$ and so 'no'	(A1)	
		or	$(2.5 \times 10^5)^2 = 2 \times 470 \times (q/m)$	(C1)	
			$(q/m) = 6.6 \times 10^7\,\text{C}\,\text{kg}^{-1}$	(M1)	
			this is less than $9.58 \times 10^7\,\text{C}\,\text{kg}^{-1}$ and so 'no'	(A1)	